SPSS MODELER AND PYTHON: MODEL COMPARISON

Student Performance Prediction

1. Model Comparison

Aspect	IBM SPSS Modeler	Jupyter Python
Dataset	Student Performance	Student Performance
Model	C&R Tree	Decision Tree Classifier
Target	Performance (Pass/Fail)	Performance (Pass/Fail)
Testing Accuracy	91.54%	83.19%
Testing Records	130	119
Correct Predictions	119	99
Wrong Prediction	11	20

2. Reason for Difference

Both models used the same dataset and the same Pass/Fail definition. The difference in accuracy is due to differences in data partitioning, model implementation and model settings in SPSS Modeler and Python.

3. Result Comparison

IBM SPSS Modeler achieved 91.54% testing accuracy, while Python achieved 83.19%. Therefore, SPSS Modeler performed better in this experiment.

4. Conclusion

Both IBM SPSS Modeler and Python successfully performed student Pass/Fail prediction. In this experiment, IBM SPSS Modeler achieved higher testing accuracy than Python. Python provides greater flexibility for coding and customization, while SPSS Modeler provides an easy visual modelling workflow.